Project: Worked examples
Title: Beam formulas (single span)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

## 1. Scope and method

Closed-form reactions, maximum moment and its position, maximum shear and maximum deflection for the standard single-span beam cases, as a quick check. Left end at x = 0; loads downward; sagging moments positive; deflection downward positive.
Roark Table 8.1 formulas in own notation. For the simply supported point load at a (b = L − a): R_A = Pb/L, M_max = Pab/L, δ_max = Pb(L² − b²)^(3/2)/(9√3 L EI) at x = √((L² − b²)/3) when a ≥ b. For the propped cantilever: R_B = 3wL/8, M_A = −wL²/8, sag M_max = 9wL²/128 at x = 5L/8.
Ref: Roark's Formulas for Stress and Strain Ch. 8, Table 8.1 — Shear, moment, slope and deflection formulas for elastic straight beams

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| case | Case | propped cantilever (fixed left, pinned right), UDL | | assumption: Fixed at the wall, propped at the free end. | any |
| L | Span | 8 | m | drawing: DRG-CA-033 rev A | 0.1 … 100 m |
| E | Modulus of elasticity | 210 | GPa | spec: BS EN 1993-1-1 3.2.6 | 1 … 500 GPa |
| I | Second moment of area | 20000 | cm^4 | reference: Fabricated section, S1 sheet | 1 … 100000000 cm^4 |
| w | Distributed load (downward) | 12 | kN/m | spec: Load schedule LS-03 | 0 … 10000 kN/m |

## 3. Assumptions

A1. Prismatic, linear elastic, Euler–Bernoulli beam; small deflections. (Roark Table 8.1 conditions.)
A2. A cantilever is fixed at the left end (x = 0) and free at the right. (Case convention.)
A3. case (Case) is taken as propped cantilever (fixed left, pinned right), UDL. (Fixed at the wall, propped at the free end.)

## 4. Working

```
[5] EI = E * I
= 210 GPa × 20000 cm⁴
= 42000 kN·m²
```
Flexural rigidity

```
[7] R_A = 5 * w * L / 8
= 5 × 12 kN/m × 8 m / 8
= 60 kN
```
Roark 8.1 2c

Fixed-end reaction

```
[8] R_B = 3 * w * L / 8
= 3 × 12 kN/m × 8 m / 8
= 36 kN
```

Propped-end reaction

```
[9] M_min = -w * L^2 / 8
= −12 kN/m × (8 m)^2 / 8
= -96 kN·m
```

Fixed-end (hogging) moment

```
[10] M_max = 9 * w * L^2 / 128
= 9 × 12 kN/m × (8 m)^2 / 128
= 54 kN·m
```

Maximum sagging moment

```
[11] x_M = 5 * L / 8
= 5 × 8 m / 8
= 5 m
```

Position of the maximum sagging moment

```
[12] V_max = 5 * w * L / 8
= 5 × 12 kN/m × 8 m / 8
= 60 kN
```

Maximum shear

```
[13] x_delta = L * (1 + sqrt(33)) / 16
= 8 m × (1 + sqrt(33)) / 16
= 3.3723 m
```

Position of the maximum deflection

```
[14] delta_max = w * L^4 / (185 * EI)
= 12 kN/m × (8 m)^4 / (185 × 42000 kN·m²)
= 6.3259 mm
```

Maximum deflection

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| R_A | 60 kN | Left reaction |
| R_B | 36 kN | Right reaction |
| M_max | 54 kN·m | Maximum sagging moment |
| x_M | 5 m | Position of M_max |
| M_min | -96 kN·m | Minimum (hogging) moment |
| V_max | 60 kN | Maximum shear |
| delta_max | 6.3259 mm | Maximum deflection |
| x_delta | 3.3723 m | Position of δ_max |

## 6. Checks (governing first)

No pass/fail checks in this calculation.

## 7. Applicability

No warnings.

## 8. Verification

Template struct.beam-formulas@1.0.0 Content hash c6e94f77a1900863 Engine calc-engine/1.0.0
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